P2 Chapter 11: Integration

Using Partial Fractions

SKILL #7: Using Partial Fractions

We saw earlier that we can split some expressions into partial fractions. This allows us to integrate some expressions with more complicated denominators.

Find $\int \frac{2}{x^2-1}\, dx$

$$\frac{2}{x^2-1} = \frac{2}{(x+1)(x-1)} = \frac{A}{x+1} + \frac{B}{x-1}$$

$$2 = A(x-1) + B(x+1)$$

$$2 = Ax - A + Bx + B$$

$$A + B = 0 \qquad B - A = 2$$

$$B = 1 \quad A = -1$$

$$\int \frac{2}{x^2-1} dx = \int -\frac{1}{x+1} + \frac{1}{x-1}\, dx$$

$$= -\mathbf{ln}|\boldsymbol{x}+\mathbf{1}| + \mathbf{ln}|\boldsymbol{x}-\mathbf{1}| + \boldsymbol{C}$$

$$= \boldsymbol{ln}\left|\frac{\boldsymbol{x}-\mathbf{1}}{\boldsymbol{x}+\mathbf{1}}\right| + \boldsymbol{C}$$

Further Examples

Find $\int \frac{x-5}{(x+1)(x-2)}\, dx$

?

Find $\int \frac{8x^2-19x+1}{(2x+1)(x-2)^2}\, dx$

?

Further Examples

Find $\int \frac{x-5}{(x+1)(x-2)}\ dx$

$$\frac{x-5}{(x+1)(x-2)} \equiv \frac{A}{x+1} + \frac{B}{x-2}$$

$A = 2\ and\ B = -1$

$$\int \frac{x-5}{(x+1)(x-2)}\ dx = \int \frac{2}{x+1} - \frac{1}{x-2} dx$$

$$= 2\ln|x+1| - \ln|x-2| + C$$

$$= \mathbf{ln}\left|\frac{(\boldsymbol{x}+\mathbf{1})^{\mathbf{2}}}{\boldsymbol{x}-\mathbf{2}}\right| + \boldsymbol{C}$$

Find $\int \frac{8x^2-19x+1}{(2x+1)(x-2)^2}\ dx$

$$\frac{8x^2-19x+1}{(2x+1)(x-2)^2}$$

$$= \frac{A}{2x+1} + \frac{B}{(x-2)^2} + \frac{C}{x-2}$$

$$A = 2, B = -1, C = 3$$

$$\int \frac{8x^2-19x+1}{(2x+1)(x-2)^2}\ dx$$

$$= \mathbf{ln}\left|(\mathbf{2}\boldsymbol{x}+\mathbf{1})(\boldsymbol{x}-\mathbf{2})^{\mathbf{3}}\right| + \frac{\mathbf{1}}{\boldsymbol{x}-\mathbf{2}} + \boldsymbol{C}$$

Test Your Understanding

Edexcel C4 June 2009 Q3

$$f(x) = \frac{4-2x}{(2x+1)(x+1)(x+3)} = \frac{A}{(2x+1)} + \frac{B}{(x+1)} + \frac{C}{(x+3)}.$$

(*a*) Find the values of the constants A, B and C. **(4)**

(*b*) (i) Hence find $\int f(x)\ dx$. **(3)**

(ii) Find $\int_0^2 f(x)\ dx$ in the form $\ln k$, where k is a constant. **(3)**

(a) ?

(b) (i) ?

(ii) ?

Test Your Understanding

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(a)

$f(x) = \frac{4-2x}{(2x+1)(x+1)(x+3)} = \frac{A}{2x+1} + \frac{B}{x+1} + \frac{C}{x+3}$		
$4-2x = A(x+1)(x+3) + B(2x+1)(x+3) + C(2x+1)(x+1)$		M1
	A method for evaluating one constant	M1
$x \to -\frac{1}{2},\ 5 = A(\frac{1}{2})(\frac{5}{2}) \Rightarrow A = 4$	any one correct constant	A1
$x \to -1,\ 6 = B(-1)(2) \Rightarrow B = -3$		
$x \to -3,\ 10 = C(-5)(-2) \Rightarrow C = 1$	all three constants correct	A1 (4)

(ii)

$\left[2\ln(2x+1) - 3\ln(x+1) + \ln(x+3)\right]_0^2$	
$= (2\ln 5 - 3\ln 3 + \ln 5) - (2\ln 1 - 3\ln 1 + \ln 3)$	M1
$= 3\ln 5 - 4\ln 3$	
$= \ln\left(\frac{5^3}{3^4}\right)$	M1
$= \ln\left(\frac{125}{81}\right)$	A1

(b) (i)

$\int\left(\frac{4}{2x+1} - \frac{3}{x+1} + \frac{1}{x+3}\right)dx$		
$= \frac{4}{2}\ln(2x+1) - 3\ln(x+1) + \ln(x+3) + C$	A1 two ln terms correct	M1 A1ft
	All three ln terms correct and "$+C$"; ft constants	A1ft (3)

SKILL #8: Integrating top-heavy algebraic fractions

$$\int \frac{x^2}{x+1}\,dx = ?$$

How would we deal with this? (the clue's in the title)

$$
\begin{array}{r|l}
 & x \quad -1 \\
\hline
x+1 & x^2 + 0x + 0 \\
 & \underline{x^2 + x} \\
 & \quad -x + 0 \\
 & \quad \underline{-x - 1} \\
 & \qquad\quad 1
\end{array}
$$

$$\int \frac{x^2}{x+1}\,dx$$

$$= \int x - 1 + \frac{1}{x+1}\,dx$$

$$= \frac{1}{2}x^2 - x + \ln|x+1| + C$$

Test Your Understanding

$$\int \frac{x}{x-1}\,dx$$

$$= \boxed{\quad ? \quad}$$

$$=$$

$$\int \frac{x^3+2}{x+1}\,dx$$

$$= \boxed{\quad ? \quad}$$

Contrast this with $\int \frac{x-1}{x}\,dx$ which can be integrated more simply:

$$\int \frac{\boldsymbol{x-1}}{\boldsymbol{x}}\boldsymbol{dx} = \int \mathbf{1} - \frac{\mathbf{1}}{\boldsymbol{x}}\,\boldsymbol{dx} = \boldsymbol{x} - \mathbf{ln}|\boldsymbol{x}| + \boldsymbol{C}$$

Test Your Understanding

$$\int \frac{x}{x-1}\,dx$$
$$= \int \frac{\boldsymbol{x-1+1}}{\boldsymbol{x-1}}\,\boldsymbol{dx} = \int \mathbf{1} + \frac{\mathbf{1}}{\boldsymbol{x-1}}\boldsymbol{dx}$$
$$= \boldsymbol{x} + \mathbf{ln}|\boldsymbol{x}-\mathbf{1}| + \boldsymbol{C}$$

$$\int \frac{x^3+2}{x+1}\,dx$$
$$= \frac{\mathbf{1}}{\mathbf{3}}\boldsymbol{x}^{\mathbf{3}} - \frac{\mathbf{1}}{\mathbf{2}}\boldsymbol{x}^{\mathbf{2}} + \boldsymbol{x}$$
$$+ \mathbf{ln}|\boldsymbol{x}+\mathbf{1}| + \boldsymbol{C}$$

Contrast this with $\int \frac{x-1}{x}\,dx$ which can be integrated more simply:

$$\int \frac{\boldsymbol{x-1}}{\boldsymbol{x}}\boldsymbol{dx} = \int \mathbf{1} - \frac{\mathbf{1}}{\boldsymbol{x}}\,\boldsymbol{dx} = \boldsymbol{x} - \mathbf{ln}|\boldsymbol{x}| + \boldsymbol{C}$$

Exercise 11.7

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Extension:

[MAT 2003 1D]

What is the exact value of the definite integral

$$\int_1^2 \frac{dx}{x + x^3}$$

Solution: ?

Exercise 11.7

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Extension:

[MAT 2003 1D]

What is the exact value of the definite integral

$$\int_1^2 \frac{dx}{x + x^3}$$

Solution: $\frac{1}{2}\ln\left(\frac{8}{5}\right)$

Homework Exercise

1 Use partial fractions to integrate the following:

a $\dfrac{3x+5}{(x+1)(x+2)}$ **b** $\dfrac{3x-1}{(2x+1)(x-2)}$ **c** $\dfrac{2x-6}{(x+3)(x-1)}$ **d** $\dfrac{3}{(2+x)(1-x)}$

2 Find the following integrals.

a $\displaystyle\int \frac{2(x^2+3x-1)}{(x+1)(2x-1)}\,dx$ **b** $\displaystyle\int \frac{x^3+2x^2+2}{x(x+1)}\,dx$ **c** $\displaystyle\int \frac{x^2}{x^2-4}\,dx$ **d** $\displaystyle\int \frac{x^2+x+2}{3-2x-x^2}\,dx$

3 $f(x) = \dfrac{4}{(2x+1)(1-2x)}$, $x \neq \pm\dfrac{1}{2}$

a Given that $f(x) = \dfrac{A}{2x+1} + \dfrac{B}{1-2x}$, find the value of the constants A and B. **(3 marks)**

b Hence find $\int f(x)\,dx$, writing your answer as a single logarithm. **(4 marks)**

c Find $\int_1^2 f(x)\,dx$, giving your answer in the form $\ln k$ where k is a rational constant. **(2 marks)**

4 $f(x) = \dfrac{17-5x}{(3+2x)(2-x)^2}$, $-\dfrac{3}{2} < x < 2$.

a Express $f(x)$ in partial fractions. **(4 marks)**

b Hence find the exact value of $\displaystyle\int_0^1 \frac{17-5x}{(3+2x)(2-x)^2}\,dx$, writing your answer in the form $a + \ln b$, where a and b are constants to be found. **(5 marks)**

Homework Exercise

5 $f(x) = \dfrac{9x^2 + 4}{9x^2 - 4}$, $x \neq \pm\dfrac{2}{3}$

a Given that $f(x) = A + \dfrac{B}{3x - 2} + \dfrac{C}{3x + 2}$, find the values of the constants A, B and C. **(4 marks)**

b Hence find the exact value of $\displaystyle\int_{-\frac{1}{3}}^{\frac{1}{3}} \frac{9x^2 + 4}{9x^2 - 4}\,dx$, writing your answer in the form $a + b\ln c$, where a, b and c are rational numbers to be found. **(5 marks)**

Problem-solving

Simplify the integral as much as possible before substituting your limits.

6 $f(x) = \dfrac{6 + 3x - x^2}{x^3 + 2x^2}$, $x > 0$

a Express $f(x)$ in partial fractions. **(4 marks)**

b Hence find the exact value of $\displaystyle\int_2^4 \frac{6 + 3x - x^2}{x^3 + 2x^2}\,dx$, writing your answer in the form $a + \ln b$, where a and b are rational numbers to be found. **(5 marks)**

7 $\dfrac{32x^2 + 4}{(4x + 1)(4x - 1)} \equiv A + \dfrac{B}{4x + 1} + \dfrac{C}{4x - 1}$

a Find the value of the constants A, B and C. **(4 marks)**

b Hence find the exact value of $\displaystyle\int_1^2 \frac{32x^2 + 4}{(4x + 1)(4x - 1)}\,dx$ writing your answer in the form $2 + k\ln m$, giving the values of the rational constants k and m. **(5 marks)**

Homework Answers

1 **a** $\ln|(x+1)^2(x+2)| + c$ **b** $\ln|(x-2)\sqrt{2x+1}| + c$

c $\ln\left|\frac{(x+3)^3}{x-1}\right| + c$ **d** $\ln\left|\frac{2+x}{1-x}\right| + c$

2 **a** $x + \ln|(x+1)^2\sqrt{2x-1}| + c$ **b** $\frac{x^2}{2} + x + \ln\left|\frac{x^2}{(x+1)^3}\right| + c$

c $x + \ln\left|\frac{x-2}{x+2}\right| + c$ **d** $-x + \ln\left|\frac{(3+x)^2}{1-x}\right| + c$

3 **a** $A = 2, B = 2$ **b** $\ln\left|\frac{2x+1}{1-2x}\right| + c$ **c** $\ln\frac{5}{9}$, so $k = \frac{5}{9}$

4 **a** $f(x) = \frac{2}{3+2x} + \frac{1}{2-x} + \frac{1}{(2-x)^2}$ **b** $\frac{1}{2} + \ln\frac{10}{3}$

5 **a** $A = 1, B = 2, C = -2$ **b** $a = \frac{2}{3}, b = -\frac{4}{3}, c = 3$

6 **a** $f(x) = \frac{3}{x^2} - \frac{1}{x+2}$ **b** $a = \frac{3}{4}, b = \frac{2}{3}$

7 **a** $f(x) = 2 - \frac{3}{4x+1} + \frac{3}{4x-1}$, so $A = 2, B = -3$ and $C = 3$

b $k = \frac{3}{4}, m = \frac{35}{27}$